Individual Contribution Report

Design and Creative Technologies

Torrens University, Australia

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Table of Contents

1. Group Details 3
2. Team Contribution Table 3
3. Appendices 5
   3.1 Appendix A – Full Experiment Log 5
   3.2 Appendix B – All-Model RMSE Visualisation 6
   3.3 Appendix C – Task 0: Data Preparation Summary 6
   3.4 Appendix D –RandomForest Cross-Task Consistency 7
4. References 9

1. Group Details

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- Source code: https://github.com/lfariabr/review-pulse
- Presentation: http://
- Video recording: http://
- Live Demo: http://

2. Team Contribution Table

I contributed to the NLP sentiment analysis project by [describe your main technical ownership: parser, preprocessing, model training, evaluation, Streamlit app, documentation, presentation]. My work focused on [specific implementation details] and helped the team meet the assessment requirements by [connect to brief: trained neural network, inference UI, evaluation, ethics, presentation].

The team divided responsibilities as follows: [summarize each member's contribution and percentage]. I estimate my contribution at [X%] because [brief justification based on code, documentation, meetings, or presentation work].

An important ethical consideration in this project is that sentiment labels can contain noise and ambiguity. Although the dataset is split into positive and negative review files, star ratings and review text may not always align. Some reviews may be mixed, sarcastic, or domain-specific, and binary classification can oversimplify real user sentiment. We addressed this by

auditing labels, documenting ambiguous cases, and reporting model limitations rather than presenting the system as universally accurate.

3. Appendices

3.1 Appendix A – Full Experiment Log

Complete record of all model variants run across Tasks 1–4. All RMSE/MAE values are on the 41-row held-out test set.

Table 1. All model variants – Tasks 1–4 – with test-set RMSE and MAE. ***Bold = best result per task.***

Task	Model	Architecture	RMSE	MAE
1	Linear	-	$8,105	$4,729
1	MLP	[64]	$5,299	$3,108
1	MLP	[64, 32]	$5,338	$2,952
1	MLP	[128, 64]	$5,500	$3,078
1	Wide & Deep	[64, 32] + linear path	$5,873	$3,283
1	**RandomForest**	**500 trees**	**$2,873**	**$2,000**
2	MLP + Z-score	[64]	$14,109	$11,878
2	MLP + Min-Max	[64]	$13,299	$10,633
2	**RF + Z-score**	**500 trees**	**$2,875**	**$2,007**
3	Linear + One-Hot	-	$15,595	$13,215
3	MLP + One-Hot	[128, 64]	$2,812	$1,773
3	RF + One-Hot	500 trees	$3,913	$2,515
3	Linear + Ordinal	-	$15,290	$12,870
3	MLP + Ordinal	[128, 64]	$6,971	$4,178
3	RF + Ordinal	500 trees	$4,587	$2,900
4	Linear	-	$15,546	$13,208
4	MLP	[128, 64]	$3,235	$2,204
4	**RandomForest**	**500 trees**	**$2,777**	**$1,902**
4	Wide & Deep	[128, 64] + linear path	$4,223	$2,647

3.2 Appendix B – All-Model RMSE Visualisation

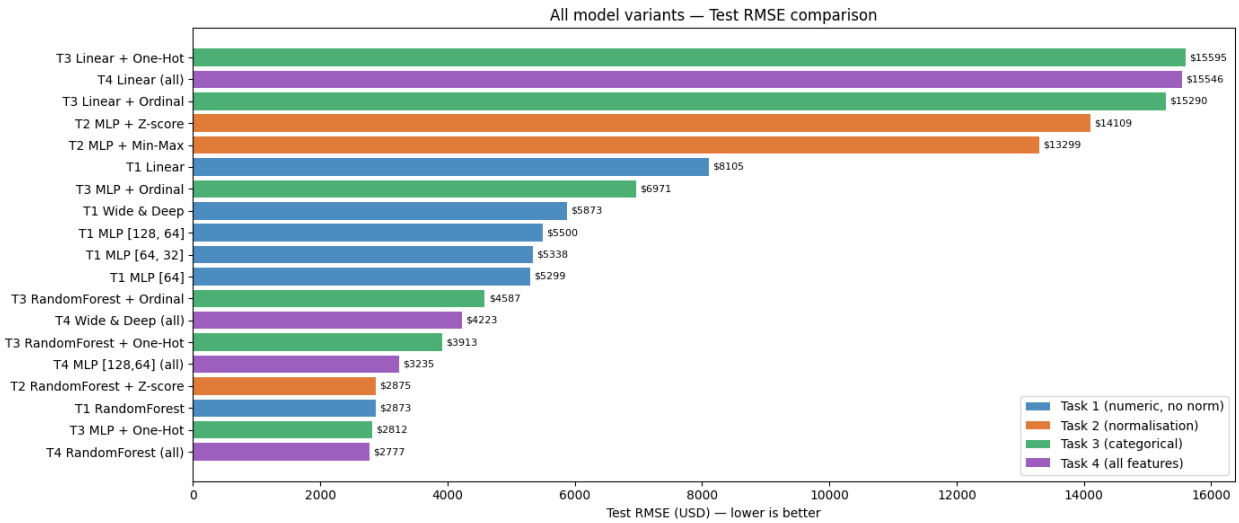

Figure 4. Horizontal bar chart of test-set RMSE for all model variants across Tasks 1–4, color-coded by task. Generated by the notebook. Lower bars = better.

3.3 Appendix C – Task 0: Data Preparation Summary

The raw UCI Autos CSV uses '?' as a placeholder for missing values. These were stored as *object dtype*, making numeric operations fail silently.

Steps applied:

1. **Convert '?' → NaN** via *pd.to_numeric(..., errors='coerce')* on all numeric columns.
2. **Drop rows where price is NaN or ≤ 0** – 4 rows removed, leaving 201 usable examples.
3. **Fill missing feature values** with the column mean – preserves rows rather than discarding them, which matters on a 205-row dataset.
4. **60/20/20 split** – *train_test_split(..., random_state=42)* → 120 train / 40 val / 41 test. All normalisation statistics computed on train only. Train label mean: $13,210. Test set mean: $13,088.

Train label mean: $13,210. Test set mean: $13,088.

3.4 Appendix D –RandomForest Cross-Task Consistency

RF is the only architecture whose RMSE is deterministic across runs and stable across all four task configurations. This table directly supports its selection as the best overall model.

Table 2. RandomForest test-set RMSE across all four tasks – showing scale-invariance and feature-set robustness.

Task	Feature set	RMSE	MAE	Notes
T1	15 numeric, no norm	$2,873	$2,000	Baseline – no encoding, no scaling
T2	15 numeric + Z-score	$2,875	$2,007	Normalisation has zero effect on RF
T3	10 categorical, one-hot (55 cols)	$3,913	$2,515	Higher RMSE – numeric signal absent
T4	**70 features (15 numeric + 15 OHE)**	**$2,777**	**$1,902**	**Best overall – RF uses all features optimally**

The rapid early convergence (epoch 5→20 drops $2,300) reflects the *make* column acting as a near-lookup-table: one-hot encoding gives each brand a dedicated weight that the model adjusts independently.

End of Appendix Section

Statement of Acknowledgment

I acknowledge that I have used the following AI tool(s) in the creation of this report:

- OpenAI ChatGPT (GPT–5)
- Anthropic Claude Sonnet 4.6

Both tools were used to assist with understanding ML concepts, structuring the technical manual, improving clarity of academic language, and supporting APA 7th referencing conventions.

Prompt examples:

1. *I implemented a TF2/Keras MLP for UCI Autos regression with a 60/20/20 split. After Z-score normalisation, RMSE jumped from $5,299 to $14,109 – worse than without normalisation. My optimizer is Adam at lr=1e-3. Can you explain the mechanism and why the same failure occurs in both Adagrad and Adam pipelines?"*
2. *"I ran a Wide & Deep model (Keras Functional API, linear + deep paths merged via layers.Add) on all 70 features. It got RMSE $4,223, worse than a plain MLP [128,64] at $3,235. Why does the Wide & Deep underperform here despite being the richer architecture?"*
3. *"Format this as APA 7th: Breiman, Leo, 2001, article titled Random Forests, journal Machine Learning, volume 45, issue 1, pages 5 to 32."*

I confirm that the use of these tools has been in accordance with the Torrens University Australia Academic Integrity Policy and TUA, Think and MDS's Position Paper on the Use of AI. I confirm that the final output is authored by me and represents my own critical thinking, analysis, and synthesis of sources. I take full responsibility for the final content of this report.

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